LIAM PELIKAN

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EDUCATION

Merrimack College — North Andover, MA
Master of Science in Computer Science, AI Concentration | **GPA: 4.0** — **Aug 2025 - May 2027**

University of Florida — Gainesville, FL
Bachelor of Science in Computer Science and Geography — **Aug 2020 - May 2024**

- **Honors:** UF Gold Scholarship, National Merit Commended Scholar, Dean's List
- **Coursework:** Machine Learning, Data Structures & Algorithms, Operating Systems, Software Engineering, Databases, Computer Networks, Linear Algebra

SKILLS

- **Languages:** Python, JavaScript, TypeScript, C++, C#, SQL, Java, HTML, CSS, Bash
- **Frameworks:** React, Next.js, Node.js, FastAPI, .NET, WPF, Tailwind CSS, LangChain
- **Data & ML:** Pandas, NumPy, Scikit-learn, OpenCV, Whisper, LLM APIs, RLHF, Feature Engineering
- **Tools:** Git, Docker, REST APIs, WebSockets, PostgreSQL, Supabase, Linux, CI/CD, FFmpeg, ArcGIS Pro

WORK EXPERIENCE

Data Annotation — Remote
AI Software Analyst — **Jan 2024 - Present**

- Review and debug AI generated Python, SQL, and TypeScript code to catch logic errors, hallucinated information, and algorithmic inefficiencies before it enters production.
- Use RLHF and pairwise ranking to clean up training datasets and flag performance regressions across foundation models.
- Design complex adversarial test cases to evaluate how well models handle tricky edge cases and multistep logic problems.
- Investigate subtle code bugs and write detailed technical reports reproducing errors to help detail model shortcomings.
- Analyze evaluation metrics across the platform to spot underlying quality trends in worker performance and identify where models break down during reasoning tasks.
- Collaborate with cross functional reviewers to standardize grading rubrics and reduce variance in subjective model feedback.

Northrop Grumman — Melbourne, FL
Software Engineering Intern, Flight Test — **May 2021 - Aug 2023**

- Wrote automated logging and Python unit tests for backend switching systems, cutting manual debugging time by 40%.
- Built simulation GUIs in WPF/.NET that let engineers compare simulated data against real hardware specs during flight test campaigns.
- Created a commit tracking tool on top of the Fisheye REST API to monitor repo activity and enforce CI/CD quality gates.
- Maintained and optimized continuous integration (CI) environments with Jenkins to streamline the deployment of software.
- Returned for three consecutive summers, working with teams across engineering disciplines on agile workflows and classified defense documentation.

PROJECTS

AI YouTube Shorts Generator — *Python, Whisper, LLM API, Docker, FFmpeg, OpenCV*

- Automated pipeline that downloads YouTube videos, transcribes them with Whisper, and uses an LLM to pick the best two minute segment for vertical short form content.
- Automatically crops videos and detects faces using OpenCV, burns in subtitles using FFmpeg, runs inside Docker, and outputs a fully rendered video file.

YouTube Content Strategist — *JavaScript, Chrome Extension (Manifest V3), LLM API*

- Chrome extension that pulls YouTube transcripts from the page and passes them to an LLM API for summarization, SEO keyword extraction, hook scoring, fact checking, and Anki flashcard generation.

MovieMatch — *Next.js, React, Supabase, PostgreSQL, Clerk Auth, Tailwind CSS*

- Full-stack movie recommendation app with swipe discovery (React Spring), real-time watchlist syncing through Supabase, OAuth via Clerk, and social matching to find shared favorites with friends.

Minesweeper — *C++, Raylib, CMake*

- Engineered a standalone C++ Minesweeper engine featuring algorithmic guess free board generation, a custom Raylib UI with persistent stat tracking, and a CMake configured Windows release.

Battleship Probabilistic Solver — *Python, Tkinter, HTML/CSS/JS, Selenium*

- Probability based solver using Monte Carlo simulations and a Tkinter UI. Built an automated Selenium web scraper to play against the best online algorithm, maintaining a positive win rate, and launched a website with my battleship algorithm.

Land Use Classification — *Python, PyTorch, Transformers, Pandas, ArcGIS*

- Architected a multimodal PyTorch pipeline to fuse visual and elevation data for training Vision Transformers.
- Engineered geospatial features from satellite and spatial datasets and fine tuned Transformer models to identify urban vs. natural land use patterns across Boulder, CO.